

Subject 001 - A Detailed Self-Report of Virtual Reality Induced Sickness

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ABSTRACT

Cybersickness is a show stopper and fortunately, the awareness of this has massively raised within the last couple of years. Still, a lot of people do not know how it feels and draw wrong or unrealistic claims from existing results. With this detailed self-report from an expert subject who is hypersensitive to cybersickness, we want to give a link to how it feels for the individual. Finally, the self-report is also set in a relation to general observations we made within experiments in the last years.

Index Terms: Human-centered concepts [Human computer interaction (HCI)]; Interaction paradigms—Virtual reality;

1 INTRODUCTION

This work does not want to create or postulate new findings regarding cybersickness. Our main goal in writing this paper is twofold, first, we would like to clarify some misconceptions around cybersickness and to raise awareness of some issues resulting from them. Second, this paper is an appeal to researchers and developers to read and apply existing results in cybersickness more carefully.

In recent years there is nothing we have been more asked about than cybersickness. Something that regularly happens is that not a few people (including scholars, researchers, and collaborators), when hearing about the phenomenon the first time in more detail are almost excited, they have a lot of questions, have heard/read about a cure already, have solutions in mind, and they do want clear and absolute answers. But they often do not want to listen— or at least are surprised— about a relativization of findings, or how effect sizes of countermeasures are often small, and neither that cybersickness is not easy to solve nor research, due to a large amount of inter-individual differences.

In empirical science effects are significant or they are not, which alone does raise issues [7]. However, without this abstraction a systematical evaluation is difficult. Variance is also nothing new in HCI and researchers have learned to deal with it. Both combined have a high impact on cybersickness research and its application to the individual. E.g., several counters have shown a significant positive effect in experiments but, simultaneously, there is still no effective method to prevent or counter cybersickness in general. Not all people just “get used to it”—while some actually do— or only have to tweak some travel parameters—although that actually helps a little bit... This means we must not forget to reverse the abstraction that was applied by the statistics when applying these results to individual humans again, which is one of the goals of this paper to make aware of.

One of the authors is (hyper)sensitive to cybersickness. To maintain a certain distance and neutrality we will refer to him as the subject in the following.

Still, he is working with VR nearly on a daily basis. In the remainder of this position paper, we want to report on his experiences resulting from this, followed by a discussion of these. We also add observations from experiments and conversations we have made

through the last years (see Section 8). We think there is a lot of value in this report when reading it with the right mindset, namely: This is not an empirical experiment and there should be no direct conclusions to be drawn from it. Nevertheless, we also think that a phenomenon as complex as cybersickness does not only need evaluation on the macro or meta scale (cf. [21,32]). The subject does not want to keep his experiences to himself and wants to contribute with this report to the micro-scale.

2 DEMOGRAPHICS, PHYSIOLOGY & ANAMNESIS

The subject is 35 years old, 1.85m tall, male, left-handed, overweight (BMI ≈ 35), and does not suffer from any visual impairments. His IPD was measured at 67mm. He suffered from heterophoria in primary school, which was cured by therapy. Also during late primary school, he suffered from stress-related migraine, which was caused by self-induced pressure or fear of failure. This was cured by autogenic training therapy, which is a collection of self-relaxation exercises. Today, the subject is healthy and does not take any medication but can still show strong body reactions when under enormous stress (see Section 5). His fitness level has decreased over recent years but he is still actively hiking and swimming. The subject does not suffer from any other form of motion sickness, such as seasickness or issues with being passively transported (motion sickness susceptibility questionnaire (MSSQ) raw score [11] = 0). His sister is also highly vulnerable to cybersickness but not so his mother. His father has not been “tested” yet.

The BMI and a low fitness-level are repeatedly reported to have a negative impact on the vulnerability to cybersickness. Furthermore, females seem to be more often affected [22, 30]. The impact of age has shown mixed results [18, 22, 27]. Having other forms of motion sicknesses is one of the strongest predictors for being also vulnerable to cybersickness [23, 30]. There are also connections between migraine and motion sicknesses reported [29].

3 USER EXPERIENCE

The subject faces a computer screen 8 to 15 hours per day. A part of these hours includes playing 2D/3D video games, which he has been doing on a regular basis since his early days. He has been using VR devices regularly (CAVEs, HMDs), for 7 years now and had sporadic experiences (CAVEs, stereoscopic displays) before. There were no dedicated occurrences of cybersickness during the first usages of virtual reality devices but these might have not been noticed because of unawareness of its existence or the types of applications exposed to.

An increasing experience with VR is often linked to habituation regarding cybersickness [13, 25, 31, 34]. Beyond this, experiences with 3D video-gaming might have a positive impact on sickness vulnerability [23].

4 OCCURRENCE

For the subject, cybersickness emerges almost exclusively within cases of continuous virtual rotation, while this is also not always the case. In the past, there have been rare single instances of VR experiences that heavily rely on virtual rotation but did not induce any sickness symptoms. The subject is not aware of any unique characteristics of those instances or applications and thus explains it with: having had a good day? While being very sensitive to virtual

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rotations the subject reports no symptoms when experiencing VR applications that are stationary, make use of teleportation, or physical travel. Therefore, he is practicing regular VR gaming without any issues usually in a standing posture.

Related work partially finds a higher impact of continuous virtual rotation in comparison to translation [15], however, there is usually an impact of both [9]. Also, a standing user seems to have shown to be more vulnerable to sickness than a seated one [19, 26].

More time the subject spends developing and testing VR applications, most of the time sitting at or in front of an office desk. In this work, there is a high focus on the design of continuous or semi-continuous locomotion methods. In these sessions, there is always at one point an occurrence of moderate to severe symptoms of cybersickness. Where moderate means, he has to stop using VR for this day but can continue working and severe means, he has to interrupt work and lay down and rest for half an hour at least. The severe case is much more regular and is often experienced multiple times during a week. To this point the subject speculates that there are much more severe instances of sickness because of the context they are occurring: He is testing and developing mainly locomotion methods, which means, first, during development the interfaces are not adult and often broken, which leads to unexpected and uncontrolled behavior, which then is also not directly quit to analyze the reasons. Second, the subject does not stop using the interfaces when symptoms start arising.

Furthermore, the subject describes the occurrence of cybersickness as event-based, and not rarely it is one single knock out event, a hard, unexpected, or uncontrollable virtual rotation that is responsible for an instant and critical increase in sickness (cf. [34]). Sometimes, cybersickness increases over time but even then he argues that it is still event-based and only the single events are not heavy enough, thus having the chance to stack.

The subject has experienced cybersickness in general and also severe instances in CAVEs as well as in HMDs. Nevertheless, the number of occurrences and their intensity is much worse with HMDs.

While this observation regarding display systems seems to be in line with general assumptions, experiments show and system parameters, such as FoV, suggest mixed results [24, 27].

Additionally, while being a teenager, there was a single instance of similar symptoms occurring while standing behind and watching another person playing a rapid first-person video game on a regular desktop computer. This never happened again and also does not translate to watching videos etc. until recently, when these symptoms repeatedly occurred again in the severe form while monitoring participants during a VR study on a control screen. The subject has no explanations for this. There is some room for speculating that it can be a form of conditioning or stress-related (see Section 5 & Section 8).

5 SYMPTOMS

When asked to give a short description of how cybersickness feels, the subject explained it is as if his whole circulatory system is going down or he got poisoned. He has never been poisoned but that is how he imagines it has to feel. When an application forces it, i.e., utilizes virtual rotation or a lot of optical flow from other sources, the symptoms arise immediately and usually force him to quit after a few minutes, sometimes even less. He admits, that after having endured hundreds of instances of cybersickness it is still not easy to precisely describe the symptoms because he is not trained to do so and it is difficult when you are sick. Over time he tried to systematically fill out simulator sickness questionnaires (SSQs) [16] and do audio recordings. These still fuzzy results are summarized in Figure 1. A lot of these assessments require further explanations which we elaborate on, in the following paragraphs.

When it comes to symptoms he has made some general observations. First, the symptoms do not all occur at the same time. Some

	moderate	severe		moderate	severe
general discomfort	●	●	fullness of the head	●	●
fatigue	●	●	blurred vision	×	×
headache	●	●	dizziness with eyes open	●	●
eye strain	●	●	dizziness with eyes closed	●	●
difficulty focusing	×	×	vertigo	●	●
salvation increasing	×	×	stomach awareness	●	●
sweating	●	●	burping	×	×
nausea	×	×	coughing	×	●
difficulty concentrating	●	●	yawning	●	×

Figure 1: Occurrences and strength of symptoms when experience either moderate or severe instances of cybersickness for the subject. The symptoms are borrowed from the SSQ [16] and extended by coughing and yawning. Crosses indicate symptoms that never occur in every instance, while lighter blue circles indicate symptoms that do not occur in every instance. The radius of a circle indicates the strength of the symptom (mild, moderate, severe). The data is not robust and based partially on: SSQs, audio memos, and recall.

symptoms are closer to the actual events that triggered the sickness (see Section 4), such as dizziness or vertigo—he notes here that he does not understand the differences between the two symptoms—and others that evolve after some time, such as headache. Second, the peak sometimes is reached up to an hour after having quit the experiences. The symptoms usually extend over 1 or 2 hours but can last over days in extreme cases (cf. [14, 24, 33]).

When thinking about the symptoms of cybersickness the SSQ is maybe an obvious point to start. And when thinking for longer times about it the questionnaire covers nearly all symptoms the subject has ever experienced. But he had to deeply analyze symptoms and reflect on himself to answer, how to describe, and finally how to name the symptoms.

This is for instance the case with **fatigue**. He always rated to feel no fatigue at all, because he first interpreted it as being physical/muscle-related or meaning being tired, while it actually feels more as being weak, which is also physical, but with a psychological component as well, comparable to the weaknesses during flu or the circulatory system shutting down. This is a very strong symptom and the main driver for his desire to lay down and rest, while it would not be represented in the questionnaire, when not also calling it fatigue.

The same is true for something that is pretty well described by the term **fullness of the head**, when not been interpreted as having a lot of problems and solutions going through your mind, but the actual presence of *something* that is filling your head but is not a headache. It is like a light overall pressure in the head, which does not hurt but still feels unpleasant.

There is also **headache**, which occurs later and especially with severe instances. It is then concentrated on the forehead and the frontal sinuses. The eyes are somehow involved in this but it is hard to say how (**eye strain**). The head-related symptoms altogether then obviously also lead to **difficulties in concentrating**.

Dizziness and **vertigo** can occur directly together with the events then triggering the sickness but then they are locally in time and not long-lasting symptoms.

Heavy **sweating**, most of the time perceived as *cold sweating* (diaphoresis) does regularly occur with his sicknesses but not always. Diaphoresis in general is also known to be induced by shock, fear, and stressful situations.

Furthermore, the subject states that there were never instances of nausea, nor vomiting. However, there is often something that can be described as **stomach awareness**, while he insists that in his

case this feeling is not located to the actual stomach but a little bit lower. Strangely, in rapidly occurring instances of sickness, such as after a really surprising and heavy loss of control about virtual rotation, he has to **cough** multiple times. It feels like instances of “dry” vomiting, without any ascension or sputum of material. This is always a clear sign to immediately quit the experience, even when other symptoms at that time had no time to arise because they will so. In cases of more slowly increasing sickness the subject multiple times finds his-self **yawning** a lot, without any presence of tiredness or other plausible reasons.

There is a final observation the subject wants to share without knowing if there is any causal link. In cases of enormous stress, such as induced by noticing to accidentally having sent an email to somebody who really should not read it, fear to miss your connecting bus in Bangkok while searching for the bus terminal, or just noticing that a paper deadline is in 1 hour and not tomorrow as expected, people often show physical reactions, close to panic. The subject experienced situations like that multiple times in his life and interestingly the body reactions are often heavy and exactly the same as with cybersickness. The extend and similarity of the reaction can be coincidental as the same symptoms do not mean the same cause. Another explanation would be that sickness of course is stressing the body [4], then the cause would be stress in both cases but the reason for the stress is another. At this point, he wants to leave out further speculation, because he is not an expert in stress and its biophysical reactions.

6 TREATMENT

The subject does not perform any special treatment other than laying down for up to an hour, putting up the legs, and closing the eyes. He prefers to drink some apple juice with sparkling water and taking some breaths of fresh air, while he has no idea if any of these have a significant effect on the recovery. When appetite comes back, eating has been shown to support recovery. He has never used any medication [25] before or after the experience because they are usually just treating the symptoms and he does not want to be constantly under medication. He also never has willingly consumed ginger before or after an exposure, which can reduce nausea [8].

7 PREVENTION & COUNTERS

The subject has researched and used a lot of known methods for preventing or countering cybersickness, where the most prominent are: sitting [19, 26], (dynamic) field of view modification [3, 10, 37], displaying a reference frame [5, 38], embodied locomotion [12, 20, 35], semi-continuous locomotion methods [1, 9], and combinations of those. While he personally attributes a positive effect to any of those, he is not sure if this is not based on a placebo effect or pure imagination, because their effect sizes tend to get lost already against his form on that day. It is not a matter of getting sick or not but whether he withstands a minute longer than last time or not, which is hard to attribute to a single factor then. Anyways, their effect increases and gets undoubtful in combination.

Furthermore, he admits that his experience could help, not in the sense of getting used to it but that he can anticipate the situations leading to sickness and avoid them [34]. An example would be to close your eyes during phases of virtual rotation.

8 DISCUSSION & OBSERVATIONS

Why does the subject think he is hypersensitive to cybersickness, or at the higher end of sickness vulnerability? It is because he has the comparison to a few hundred participants—none of the studies actively was trying to induce sickness—and people just experiencing VR. Even when taking into account that participants often do not want to openly show weaknesses—especially men, which unfortunately still make a predominant percentage of participants—there was never one showing similar reactions that strong, except for his

sister. Still, he observed a minority of people also showing strong reactions that had to rest some minutes after the exposure. Nevertheless, it is important to note that this minority is not alone. Indeed, our observations are that in VR experiences with continuous locomotion—especially when it includes virtual rotation—that reach or exceed the one-hour-exposure time the part of the population not showing any effects of sickness is the minority (cf. [6]).

When it comes to the correlation of individual factors, such as gender, VR experience, age, fitness level, MSSQ score, video game experience, etc. [17, 23, 28], we think they are important for finding the actual reasons for sickness but their variance between individuals is much too high and their interaction unknown for being used the other way around, a vulnerability prediction of an individual.

Except for coughing and yawning, the symptoms reported by the subject, are covered by the SSQ [16]. However, it can take a significant effort in time and serious analysis for a layman to make symptoms, such present that they get distinguishable and describable. We doubt that this is the case for a regular study participant. In the absence of alternatives, the scale still does its job to make cybersickness comparable between either the same individual or populations but should in our opinion only carefully be used for further analysis of the sub-scales, oculomotor, disorientation, and nausea, in a virtual environment. The two added symptoms, coughing and yawning, in our observation are pretty unique and give a lot of room for speculation. For our subject, those are very connected to the sickness but without any other supporting data, we want to leave it with the reporting. The same is true with a possible connection to regular stress symptoms and stress vulnerability, which are interesting and should find some attention in future research.

The subject is very confident that at least in his case, the sickness has not much to do with factors that are generally giving in VR, such as light emission or vergence-accommodation conflict as he can consume VR experiences such as Beat Saber for long periods without any effects but being physically exhausted. It seems to be much more a question of being in control or not. But why is he not vulnerable to other forms of motion sickness then? Nevertheless, control does also play an important role in the observation that some people that do not get familiar with leaning-based interfaces tend to report discomfort and sickness [36], while the overall effect of embodied interfaces seems to be beneficial [12, 20, 35], which also makes sense in presence of additional and appropriate vestibular cues. This could be an indicator for “the actual feeling of being in control” being a more important factor than a smaller sensory cue conflict, which would support postural instability theory [26].

In line with the subject’s feelings, we observed most of the counters, such as described in Section 7 (see also [2]), to have generally a positive effect, which supports related work. However, their effectiveness often is overestimated by undergrads and researchers without experience in the observation of participants affected by sickness. Users do still get sick when sitting [36], to name just one example. Today, in the best case counters are extending the period of use for people that are affected by cybersickness. Additionally, this might get only observable when exposure times in experiments are closer to actual exposure times in reality (see [21, 32, 36]).

Running these experiments is difficult. And it is difficult to then run multiple conditions per participant in a row. But this is not a good idea when doing cybersickness research anyways because of potential carry-over effects. Furthermore, research questions regarding sickness are nothing that should be, by the way, added to experiments with a completely other focus, when they want to have a serious impact on the understanding of cybersickness. This is because investigating cybersickness often requires costlier study designs (see above) and in particular more participants, than studies with a pure focus on interaction. Mixing both might quickly end in bad compromises. This is not a call to stop using SSQs in sickness unrelated user studies, it is an important control parameter. It is

a call to stop drawing strong claims regarding cybersickness from these results.

9 CONCLUSION

The awareness of cybersickness being an actual problem in general and not only for a minority of people has risen in the last couple of years. Still, there is also a lot of trivialization and unrealistic expectations present. With this self-report and the discussion of it we hope to have added some more awareness to this. The subject's experiences and history in many parts are in line with current findings, where other parts contradict. We claim that this is the case for most people that are vulnerable to cybersickness and that there is not a simple pattern.

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